Lecture 7

States, Liouville's theorem and the second answer

Objectives. *Lectures 5 and 6 moved the observables along Hamiltonian flows. This lecture moves the states. A state need not be a point: incomplete knowledge is a probability distribution on phase space. To move one we need what the flow does to volume, and the answer — Hamiltonian flows preserve it — is Liouville's theorem. States then move by the Liouville equation, and moving the states or moving the observables gives the same predictions. Volume also settles the question Lecture 1 asked about attractors. The lecture closes with the second answer to the course's question: a classical mechanical system is a Poisson algebra of observables, a distinguished element H, and a convex set of states.*

7.1 States

Throughout, $\mathcal{M} \subseteq \mathbb{R}^{2n}$ is open with coordinates $x = (q, p)$, observables are the real C^∞ functions on $\mathcal{M}$ (Definition 5.1), and $H \in C^\infty(\mathcal{M})$ has flow φ_t, the flow of $\dot{x} = J\nabla H$ (4.9). Integrals are against Lebesgue measure $\mathrm{d}x = \mathrm{d}q_1 \cdots \mathrm{d}q_n \, \mathrm{d}p_1 \cdots \mathrm{d}p_n$, and vol is its volume.

Until now a state has been a point of $\mathcal{M}$ (Definition 1.1): the initial data that determine the motion. An experimenter rarely has them. What a repeatable preparation delivers is a *distribution* of initial data, and every prediction is then a distribution of readings.

Definition 7.1 (Density, state). A *density* on $\mathcal{M}$ is a continuous $\rho\colon \mathcal{M} \to [0, \infty)$ with $\int_{\mathcal{M}} \rho(x)\,\mathrm{d}x = 1$. A *state* is a finite convex combination

$$\sigma = \sum_{k=1}^{r} \lambda_k \rho_k + \sum_{j=1}^{s} \mu_j \delta_{z_j}, \qquad \lambda_k, \mu_j \geq 0, \quad \sum_k \lambda_k + \sum_j \mu_j = 1, \tag{7.1}$$

of densities ρ_k and of points $z_j \in \mathcal{M}$. It acts on observables by

$$\langle f \rangle_\sigma = \sum_k \lambda_k \int_{\mathcal{M}} f \rho_k \,\mathrm{d}x + \sum_j \mu_j f(z_j), \tag{7.2}$$

the *expectation* of f, whenever the integrals converge absolutely. The states form a set $\mathcal{S}(\mathcal{M})$; a state of the form δ_z is a *Dirac state*.

Every bounded observable has an expectation. For a growing f the expectation may be infinite, and the state then predicts no mean for that device. The functional (7.2) is linear, positive and normalised, and $\mathcal{S}(\mathcal{M})$ is *convex* by construction: a combination of two sums of the form (7.1) is another such sum (Figure 7.1).

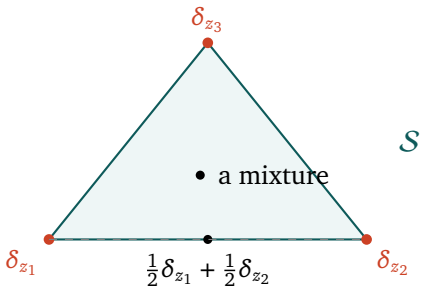


Figure 7.1: The mixtures of three Dirac states form a triangle: the vertices are the Dirac states and every other point is a mixture. Proposition 7.2 identifies vertex, dispersion-free and Dirac on any $\mathcal{M}$.

The reading of an observable

A state predicts more than the number $\langle f \rangle_\sigma$: the probability that a reading of f falls below λ is $\langle \mathbf{1}_{\{f<\lambda\}} \rangle_\sigma$, and letting λ run gives the whole distribution of the reading (Figure 7.2). Indicators jump, so they are not observables — Lecture 5 set them aside for exactly this moment — but a state integrates them anyway, and $\langle \mathbf{1}_B \rangle_\sigma$ is the probability of finding the system in the region B.

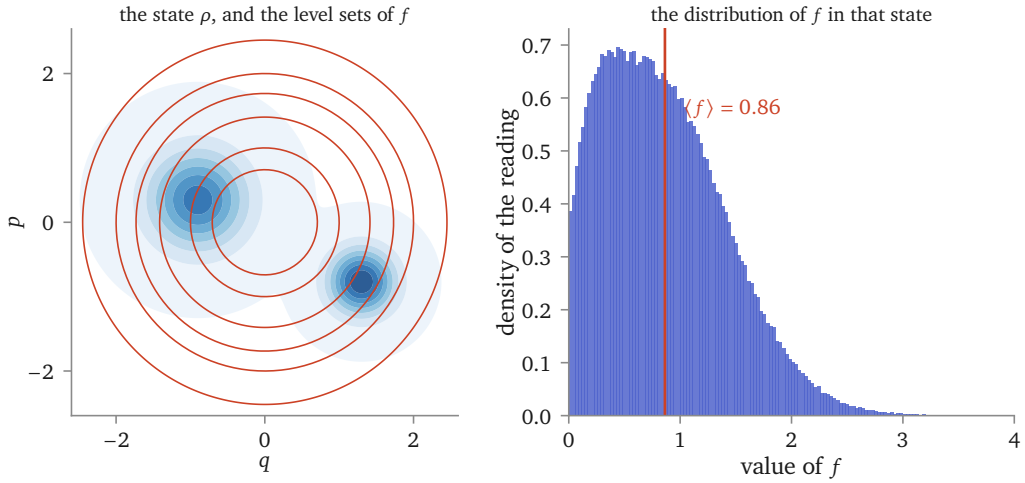


Figure 7.2: Left: a state (a mixture of two Gaussians) with the level sets of $f = \frac{1}{2}(q^2 + p^2)$ over it. Right: the distribution of the reading of f in that state, with $\langle f \rangle$ marked.

Pure states

Proposition 7.2 (Pure states)**.** *For $\sigma \in \mathcal{S}(\mathcal{M})$ the following are equivalent.*

1. *$\sigma = \delta_z$ for some $z \in \mathcal{M}$.*
2. *Each of the $2n$ coordinates $x_a = q_1, \dots, p_n$ has $\langle x_a^2 \rangle_\sigma < \infty$ and variance $\mathrm{Var}_\sigma(x_a) = \langle x_a^2 \rangle_\sigma - \langle x_a \rangle_\sigma^2 = 0$.*
3. *σ is an extreme point of $\mathcal{S}(\mathcal{M})$: it is not $\alpha\sigma_1 + (1-\alpha)\sigma_2$ with $0 < \alpha < 1$ and $\sigma_1 \neq \sigma_2$.*

Such a state is called pure*, and every other state is* mixed*.*

Proof. One observation is used twice. $(*)$ *If $g \geq 0$ is continuous on $\mathcal{M}$, $g(x) > 0$ for $x \neq c$,*

and $\langle g\rangle_\sigma = 0$*, then* $c \in \mathcal{M}$ *and* $\sigma = \delta_c$. Indeed every term of (7.2) applied to g vanishes. A density component would give $\int \rho_k g = 0$ with a non-negative continuous integrand, so ρ_k would vanish off the point c and, by continuity, everywhere — against $\int \rho_k = 1$. So every $\lambda_k = 0$, some $\mu_j > 0$, and $\mu_j g(z_j) = 0$ leaves $z_j = c$ whenever $\mu_j > 0$.

(1)⇒(2). $\langle f^2\rangle_{\delta_z} = f(z)^2 = \langle f\rangle_{\delta_z}^2$.

(2)⇒(1). Each $\langle x_a\rangle_\sigma = c_a$ is finite, since $|x_a| \le 1 + x_a^2$. Then $\langle |x-c|^2\rangle_\sigma = \sum_a \mathrm{Var}_\sigma(x_a) = 0$, and $(*)$ with $g = |x-c|^2$ gives $\sigma = \delta_c$.

(1)⇒(3). If $\delta_z = \alpha\sigma_1 + (1-\alpha)\sigma_2$ with $0 < \alpha < 1$, take the bounded observable $g = |x-z|^2/(1+|x-z|^2)$. Then $0 = g(z) = \alpha\langle g\rangle_{\sigma_1} + (1-\alpha)\langle g\rangle_{\sigma_2}$ with both expectations non-negative, so both vanish, and $(*)$ gives $\sigma_1 = \sigma_2 = \delta_z$.

(3)⇒(1). Let σ be no Dirac state, written as in (7.1) with positive weights. If some $\mu_j \in (0,1)$ then $\sigma = \mu_j\delta_{z_j} + (1-\mu_j)\sigma'$, and $\sigma' \neq \delta_{z_j}$ because otherwise $\sigma = \delta_{z_j}$: a non-trivial mixture. No μ_j is 1, since that is the excluded case. So every $\mu_j = 0$ and $\sigma = \sum_k \lambda_k\rho_k$ is itself a density ρ, a convex combination of densities being one. Put $\chi = e^{q_1}/(1+e^{q_1})$, continuous with $0 < \chi < 1$, and $\alpha = \int \chi\rho \in (0,1)$; then $\rho = \alpha\rho_1 + (1-\alpha)\rho_2$ with $\rho_1 = \chi\rho/\alpha$ and $\rho_2 = (1-\chi)\rho/(1-\alpha)$, both densities. They differ: $\rho_1 = \rho_2$ would force $\chi \equiv \alpha$ on the non-empty open set $\{\rho > 0\}$, which a strictly increasing function of q_1 is not. ∎

A pure state is thus a point of $\mathcal{M}$ seen three ways: perfect knowledge; every position and momentum definite; not preparable by tossing a coin between two other states.

7.2 Liouville's theorem

To move states we must first know what the flow does to volume. Lecture 6 answered this through the symplectic group: $D\varphi_t \in \mathrm{Sp}(2n)$ and $\det D\varphi_t = \pm 1$ by Proposition 6.14, hence $+1$, since it is continuous in t and equals 1 at $t = 0$. The proof below needs no symplectic linear algebra and applies to any vector field — which the damped pendulum needs, since its flow is not Hamiltonian.

Lemma 7.3 (Abel–Liouville formula)**.** *Let* $A\colon I \to \mathbb{R}^{N\times N}$ *be continuous and let* $M\colon I \to \mathbb{R}^{N\times N}$ *solve* $\dot M = A(t)M$*. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\det M(t) = \operatorname{tr}A(t)\,\det M(t), \qquad \text{so} \qquad \det M(t) = \det M(t_0)\exp\int_{t_0}^{t}\operatorname{tr}A(\tau)\,\mathrm{d}\tau. \tag{7.3}$$

Proof. The determinant is multilinear in the columns $m_1, \dots, m_N$ of M, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\det M = \sum_{j=1}^{N}\det\bigl[m_1,\dots,\dot m_j,\dots,m_N\bigr] = \sum_{j=1}^{N}\det\bigl[m_1,\dots,Am_j,\dots,m_N\bigr].$$

The last sum equals $\operatorname{tr}A\cdot\det M$. Indeed, when M is invertible write $Am_j = \sum_i B_{ij}m_i$ with $B = M^{-1}AM$; only the $i = j$ term survives in the j-th determinant, which is therefore $B_{jj}\det M$, and $\sum_j B_{jj} = \operatorname{tr}B = \operatorname{tr}A$. Both sides are polynomials in the entries of M and A, and they agree on the invertible matrices, which are dense, so they agree everywhere. Finally $\mathrm{d}/\mathrm{d}t\bigl(\det M\, e^{-\int \operatorname{tr}A}\bigr) = 0$. ∎

Theorem 7.4 (Liouville's theorem · *proved modulo T2, T5 (cited)*)**.** *Let* X *be a* C^∞ *vector*

field on $\mathcal{M}$ with flow φ_t. Then

$$\det D\varphi_t(x) = \exp \int_0^t \operatorname{div} X(\varphi_\tau x)\, \mathrm{d}\tau \tag{7.4}$$

wherever $\varphi_t x$ is defined. In particular, if $\operatorname{div} X \equiv 0$ then $\det D\varphi_t \equiv 1$ and $\operatorname{vol}(\varphi_t(D)) = \operatorname{vol}(D)$ for every bounded measurable D with $\overline{D} \subseteq \mathcal{M}$ on which φ_t is defined. For $X = X_H = J\nabla H$ this hypothesis holds:

$$\operatorname{div} X_H = \sum_{i=1}^{n} \left(\frac{\partial}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial}{\partial p_i}\frac{\partial H}{\partial q_i} \right) = 0, \tag{7.5}$$

by equality of mixed partial derivatives.

Proof. Fix x and put $M(t) = D\varphi_t(x)$. By T5 (Theorem 1.7) M is C^1 in t on the interval I_x where the motion exists (Theorem 1.5) and solves the variational equation $\dot{M} = DX(\varphi_t x)M$ with $M(0) = I$. Its trace is $\operatorname{tr} DX = \operatorname{div} X$, so (7.3) gives (7.4). When $\operatorname{div} X \equiv 0$ the exponent vanishes, and the change of variables $y = \varphi_t(x)$ gives $\operatorname{vol}(\varphi_t(D)) = \int_D \left|\det D\varphi_t(x)\right| \mathrm{d}x = \operatorname{vol}(D)$. ∎

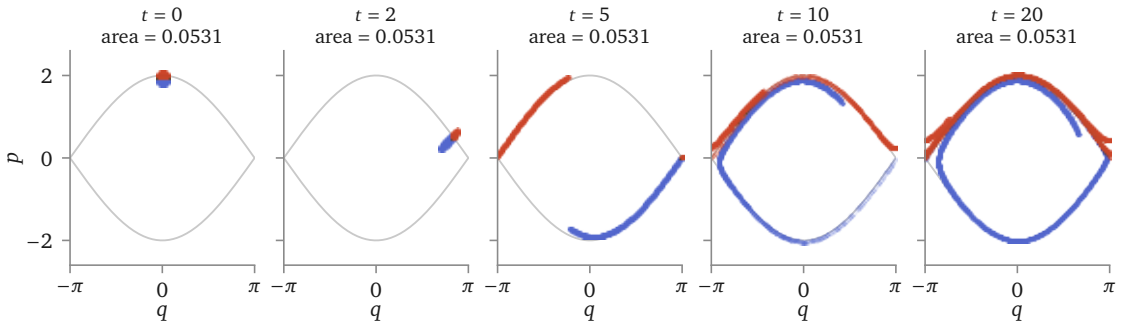


Figure 7.3: A disc of initial conditions straddling the pendulum's separatrix $H = 2$, drawn on the cylinder at $t = 0, 2, 5, 10, 20$; blue librates, red rotates. The shape is destroyed and the area is not: the number under each panel is the area of the transported boundary, computed with a non-symplectic integrator so that its constancy is a measurement.

Figure 7.3 is the picture to keep. A blob of initial conditions is stretched into a filament so fine that at any fixed resolution it soon looks spread out — and its area is what it was. Each motion keeps its energy, so none crosses the separatrix: the librating part and the rotating part never mix.

The contrast: friction

For the damped pendulum (1.16), written as $\dot{q} = p$, $\dot{p} = -\sin q - \gamma p$, the divergence is $-\gamma$, so (7.4) gives $\det D\varphi_t = e^{-\gamma t}$ exactly: every volume shrinks at the same exponential rate, whatever the orbits do (Figure 7.4). This is the quantitative form of the contrast drawn in § 1.6, and § 7.4 turns it into a theorem.

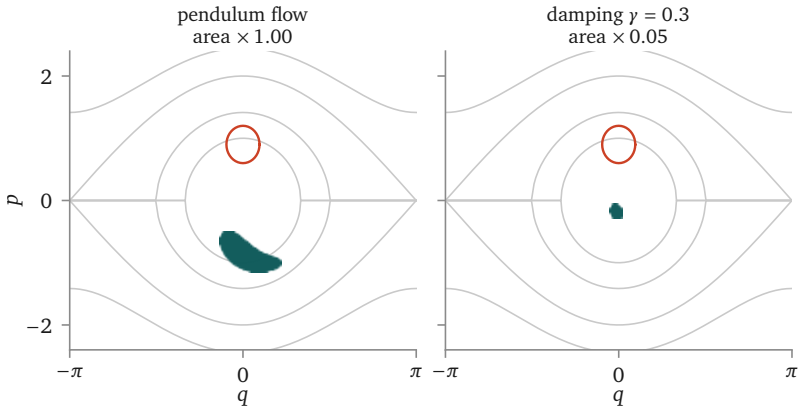


Figure 7.4: A disc of initial conditions inside the separatrix after $t = 10$: under the pendulum's flow, and under the damped pendulum with $\gamma = 0.3$; the red circle is where it started. The damped field has $\operatorname{div} X = -\gamma$, so by (7.4) its area is multiplied by e^{-3}.

7.3 The two pictures

From here on H does not depend on t and its flow is complete (Definition 1.8), as for $\frac{1}{2}|p|^2 + V(q)$ with $V \in C^\infty(\mathbb{R}^n)$ bounded below (Proposition 1.9).

Lecture 5 moved the observables and kept the state fixed: $f_t = f \circ \varphi_t$ and $\partial_t f_t = \{f_t, H\}$ (Theorem A, Theorem 5.14). That is the *Heisenberg picture*. The alternative is to keep the observables and move the state.

Transporting a state

A Dirac state moves with the point it sits at: after time t the system prepared at z is at $\varphi_t z$, so δ_z becomes $\delta_{\varphi_t z}$. For a density the rule must be found. The probability of finding the system in a region D at time t is the probability that it started in $\varphi_{-t}(D)$, so the transported density ρ_t has to satisfy $\int_D \rho_t = \int_{\varphi_{-t}(D)} \rho$ for every D. Change variables by $x = \varphi_{-t}(y)$, whose Jacobian is 1 by Theorem 7.4:

$$\int_{\varphi_{-t}(D)} \rho(x)\,\mathrm{d}x = \int_D \rho\big(\varphi_{-t}(y)\big)\mathrm{d}y.$$

As this holds for every D and both integrands are continuous,

$$\rho_t = \rho \circ \varphi_{-t}. \tag{7.6}$$

Volume preservation is not decoration here: without it the right-hand side would carry a Jacobian factor, and (7.6) would be false. Transporting each component of (7.1) this way gives σ_t, whose densities are again continuous, non-negative and of integral 1: the transported state is again a state.

The sign in (7.6) has a reason. Observables are read where they are, so they are composed with φ_t; a density is *carried along*, so its value at a point now is its old value at the point that flowed *into* it, and it is composed with φ_{-t}.

Theorem 7.5 (The Liouville equation · *proved modulo T2, T5 (cited)*)**.** *Let ρ be a density of*

class C^∞. Then $(t,x) \mapsto \rho_t(x)$ is C^∞ and

$$\frac{\partial \rho_t}{\partial t} = \{H, \rho_t\}\,. \tag{7.7}$$

Proof. Put $u_s = \rho \circ \varphi_s$. By Theorem A it is C^∞ in (s,x) with $\partial_s u_s = \{u_s, H\}$. Since $\rho_t = u_{-t}$, the chain rule gives $\partial_t \rho_t = -(\partial_s u_s)\big|_{s=-t} = -\{\rho_t, H\} = \{H, \rho_t\}$. ∎

So the two sides of mechanics are generated by the two orders of the same bracket: $\{\cdot, H\}$ moves observables, $\{H, \cdot\}$ moves states, and they are exact negatives.

Which C^∞ densities do not move? If $\rho_t = \rho$ for every t then (7.7) gives

$$\{H, \rho\} = 0; \tag{7.8}$$

conversely (7.8) makes ρ conserved (Proposition 5.10), constant along every motion, so $\rho \circ \varphi_{-t} = \rho$. The densities at rest are the conserved observables that happen to be densities.

The two pictures must make the same predictions.

Theorem 7.6 (The pictures agree · *proved modulo T2, T5 (cited)*)**.** *Let σ be a state and $t \in \mathbb{R}$. Then for every observable f with $\langle |f| \rangle_{\sigma_t} < \infty$ — equivalently, by the same change of variables, $\langle |f \circ \varphi_t| \rangle_\sigma < \infty$ —*

$$\langle f \circ \varphi_t \rangle_\sigma = \langle f \rangle_{\sigma_t}. \tag{7.9}$$

Proof. For a Dirac component both sides are $f(\varphi_t z)$. For a density component, change variables by $y = \varphi_t(x)$, whose Jacobian is 1:

$$\int f(\varphi_t x)\rho(x)\,\mathrm{d}x = \int f(y)\,\rho(\varphi_{-t} y)\,\mathrm{d}y = \int f \rho_t\,\mathrm{d}y.$$ ∎

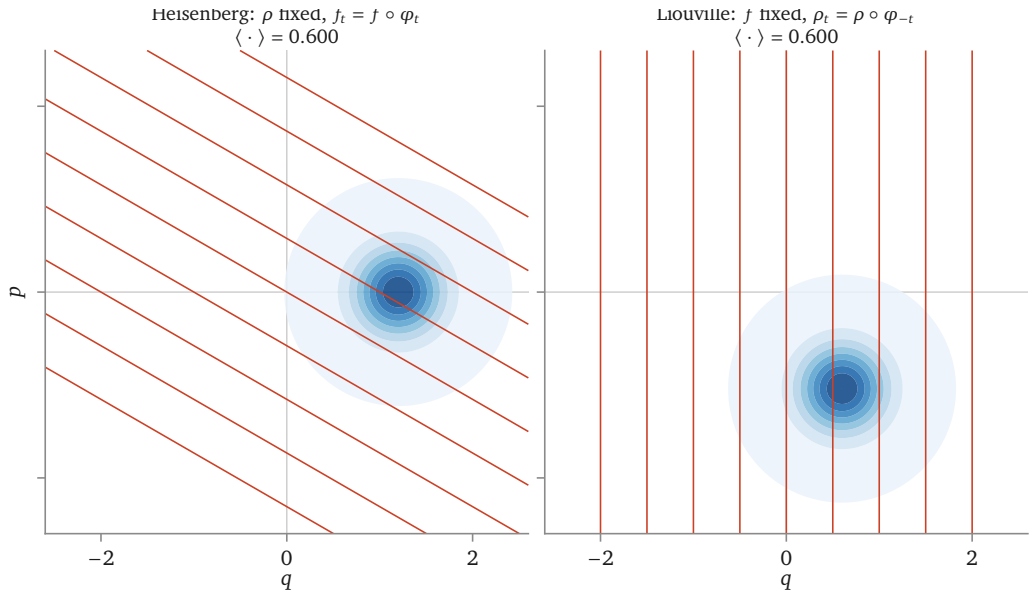


Figure 7.5: The two pictures of one number, for the oscillator and $f = q$. Left: the state stays and the level sets of $f_t = f \circ \varphi_t$ turn. Right: the level sets of f stay and the state turns the other way. The expectation is the same, and it is $1.2 \cos t$.

Adjointness

There is a second, differential way to see (7.9), and it is the one that survives into quantum mechanics.

Proposition 7.7 (The bracket is a divergence; the integral is cyclic)**.** *For $f, g, h \in C^\infty(\mathcal{M})$,*

$$\{f, g\} = \nabla f \cdot J\nabla g = \operatorname{div}\big(f\, X_g\big). \tag{7.10}$$

If moreover fh has compact support in $\mathcal{M}$, then

$$\int_{\mathcal{M}} \{f, g\}\; h \,\mathrm{d}x = \int_{\mathcal{M}} f\; \{g, h\}\; \mathrm{d}x. \tag{7.11}$$

Proof. $\nabla f \cdot J\nabla g = \sum_i (\partial_{q_i} f\, \partial_{p_i} g - \partial_{p_i} f\, \partial_{q_i} g) = \{f, g\}$, and $\operatorname{div}(fX_g) = \nabla f \cdot X_g + f \operatorname{div} X_g = \{f, g\}$ by (7.5). For a compactly supported a the field aX_g extends by zero to a C^∞ field on $\mathbb{R}^{2n}$ of compact support, and each term $\int \partial_b (aX_g)_b$ vanishes on integrating that one variable first: $\int \{a, g\}\, \mathrm{d}x = \int \operatorname{div}(aX_g)\mathrm{d}x = 0$, with no boundary regularity of $\mathcal{M}$ needed. Apply this to $a = fh$ and use Leibniz (Proposition 5.3): $0 = \int \{fh, g\} = \int \{f, g\}\; h + \int f\; \{h, g\}$, which is (7.11). ∎

Some decay is genuinely needed: for $f = q$, $g = p$, $h = 1$ on $\mathcal{M} = \mathbb{R}^2$ the left side of (7.11) is $\int 1 \,\mathrm{d}x = \infty$ and the right side is 0. With it, differentiate the two sides of (7.9) at $t = 0$, for a compactly supported f and a C^∞ density ρ. The left gives $\int \{f, H\}\; \rho$ by Theorem A; the right gives $\int f\, \partial_t \rho_t|_{t=0} = \int f\; \{H, \rho\}$ by (7.7). That the two agree is precisely (7.11) with $g = H$ and $h = \rho$.

In words: with respect to the pairing $\langle \rho, f \rangle = \int \rho f$, the generator $\{\cdot, H\}$ of the Heisenberg equation and the generator $\{H, \cdot\}$ of the Liouville equation are *adjoint* to one another. The Liouville equation is the adjoint of the Heisenberg equation — the classical ancestor of the relation between the Heisenberg and von Neumann equations.

7.4 No attractors

Lecture 1 asked what forbids attractors in mechanics. Volume settles the question, for attractors of every kind.

Definition 7.8 (Trapping region, attractor)**.** A *trapping region* for a flow is a non-empty bounded open $U \subseteq \mathcal{M}$ with $\overline{U} \subseteq \mathcal{M}$ on which the flow is defined for all $t \geq 0$ and satisfies $\varphi_t(U) \subseteq U$. The set it traps is $\Lambda = \bigcap_{t \geq 0} \varphi_t(U)$, and an *attractor* is a set of volume zero trapped by some trapping region. Sinks, attracting limit cycles and strange attractors are attractors in this sense.

Theorem 7.9 (Nothing is attracted · *proved modulo T2, T5 (cited)*)**.** *Let U be a trapping region for X_H. Then* $\operatorname{vol}(\Lambda) = \operatorname{vol}(U) > 0$. *In particular a Hamiltonian flow has no attractor.*

Proof. For $t \geq s \geq 0$ we have $\varphi_t(U) = \varphi_s(\varphi_{t-s}(U)) \subseteq \varphi_s(U)$, so the open sets $\varphi_t(U)$ decrease as t grows, and Λ is already the intersection over the integers $m \geq 0$. Each $\varphi_m(U)$ is a subset of U with $\operatorname{vol}(\varphi_m(U)) = \operatorname{vol}(U) < \infty$ by Theorem 7.4, so $\operatorname{vol}\big(U \setminus \varphi_m(U)\big) = 0$; and $U \setminus \Lambda = \bigcup_{m \geq 0} \big(U \setminus \varphi_m(U)\big)$ is a countable union of sets of volume zero, hence of volume zero.[1]

[1] This is the one place where volume is used beyond finite additivity. A countable union of sets of volume zero has volume zero, by covering the m-th by open sets of total volume $\varepsilon 2^{-m}$.

So $\mathrm{vol}(\Lambda) = \mathrm{vol}(U)$, which is positive because U is open and non-empty. ∎

"Trapping" is not a technicality. The flow of $H = -qp$ is $\varphi_t(q,p) = (e^{-t}q, e^t p)$, which preserves area while every point approaches the null line $\{q = 0\}$. There is no trapping region: the orbits run off along p as they close in. Volume stops a flow compressing a region onto a thin set, not stretching one out along one.

This closes the question of § 1.6. The damped pendulum has an attractor because its flow contracts volume at the rate γ; a Hamiltonian flow contracts nothing, so Theorem 7.9 rules out the sink, the attracting limit cycle and the strange attractor together. What a Hamiltonian flow can do instead is Figure 7.3: spread a blob out without shrinking it.

7.5 What a mechanical system is: the second answer

The question of Lecture 1 now has its second answer.

Definition 7.10 (Classical mechanical system, second answer)**.** A *classical mechanical system* consists of

1. a Poisson algebra $\mathcal{A}$ of observables (Definition 5.7);
2. a distinguished element $H \in \mathcal{A}$, the Hamiltonian;
3. a convex set $\mathcal{S}$ of states. A state assigns to an observable its expectation $\langle f \rangle_\sigma$, linearly and positively, with $\langle 1 \rangle_\sigma = 1$ — on the observables it can read, which for Definition 7.1 is not all of $\mathcal{A}$: the Gaussian state on $\mathbb{R}^2$ gives no value to $e^{q^2+p^2}$. Its extreme points are the pure states.

The dynamics is generated by H: observables evolve by $\dot f = \{f, H\}$ and states by the adjoint equation, which must carry $\mathcal{S}$ into itself, and the two give the same expectations, $\langle f_t \rangle_\sigma = \langle f \rangle_{\sigma_t}$.

Each clause has been proved for $\mathcal{A} = C^\infty(\mathcal{M})$ and the states of Definition 7.1: the algebra is a Poisson algebra (Theorem 5.5 and Proposition 5.3); the flow acts on it by automorphisms (Theorem B, Theorem 6.4); the states are convex with the Dirac states as extreme points (Proposition 7.2); for a complete flow the transported state is again a state (§ 7.3); and the two pictures agree (Theorem 7.6). The pendulum passes through all three layers:

	Lagrange	**Hamilton**	**Poisson**
Arena	$Q = S^1$	$\mathbb{R}^2$, the angle read modulo 2π (§ 1.5)	$\mathcal{A} = C^\infty(\mathcal{M})$ with $\{\cdot,\cdot\}$
Object	the curve $q(t)$	the point (q,p)	the state σ
Law	$\mathcal{L} = \frac{1}{2}\dot q^2 + \cos q - 1$ stationary	$\dot x = J\nabla H$, $H = \frac{1}{2}p^2 + 1 - \cos q$	$\dot f = \{f,H\}$, $\partial_t \rho = \{H, \rho\}$
Reading	—	$f(q,p)$	$\langle f \rangle_\sigma$

The three columns are the same mechanics. Left to right costs the manifest covariance of $\mathcal{L}$ and buys a state that is a point, then a state that need not be; right to left, Proposition 7.2 takes the Dirac states back to points, and Theorem 4.5 takes their orbits back to the curves.

Summary.

- *Handed on:* states, finite mixtures of densities and points, with their expectations (Definition 7.1); pure = Dirac = dispersion-free = extreme (Proposition 7.2); Liouville's theorem, $\det D\varphi_t = \exp \int_0^t \mathrm{div}\, X$ and $\mathrm{div}\, X_H = 0$ (Theorem 7.4); states move by $\rho_t = \rho \circ \varphi_{-t}$ and $\partial_t \rho_t = \{H, \rho_t\}$ (Theorem 7.5); the two pictures agree (Theorem 7.6), and the Liouville

equation is the adjoint of the Heisenberg equation (Proposition 7.7); a Hamiltonian flow has no attractor (Theorem 7.9), which answers the question of § 1.6.

- *The second answer:* a classical mechanical system is a Poisson algebra of observables, a distinguished H, and a convex set of states (Definition 7.10).
- *Next:* Lecture 8 builds the maps that preserve all this structure, the canonical maps, from a single function.

References

Faddeev and Yakubovskii, *Lectures on Quantum Mechanics for Mathematics Students*, §§2–3 (pp. 6–15): states as distributions, pure states, and the Jacobian proof of Liouville's theorem. Their bracket is the negative of (5.1), so their observable equation $\dot{f} = \{H, f\}$ and their density equation $\dot{\rho} = -\{H, \rho\}$ are our (5.2) and (7.7), and their $\rho(g_{-t}x)$ is our (7.6). Neumaier and Westra, *Classical and Quantum Mechanics via Lie Algebras*, §§1.7–1.8 (pp. 13–17): the Liouville equation as the adjoint of the Heisenberg equation. Arnold, *Mathematical Methods of Classical Mechanics*, §16 (pp. 68–70): Liouville's theorem by the divergence.